

20260923 Quiz

for (MATH200) Differential Equations

Jungwoo Kim

Department of Mathematics

Pohang University of Science and Technology

Problem 2.4.1. Determine, without solving the problem, an interval in which the solution of the initial value problem

$$(t - 5)y' + (\ln t)y = 2t, \quad y(1) = 2,$$

is certain to exist.

Solution. Dividing by $t - 5$ gives the standard form

$$y' + \frac{\ln t}{t - 5}y = \frac{2t}{t - 5}.$$

The coefficients are continuous where $t > 0$ and $t \neq 5$. Thus the largest open interval containing the initial point $t = 1$ on which both coefficients are continuous is

$$(0, 5).$$

By Theorem 2.4.1, the solution is certain to exist throughout this interval.

□

Problem 2.4.2. Determine, without solving the problem, an interval in which the solution of the initial value problem

$$y' + (\tan t)y = \sin t, \quad y(2) = 0,$$

is certain to exist.

Solution. Here

$$p(t) = \tan t, \quad g(t) = \sin t.$$

The function g is continuous on $\mathbb{R}$, while p is discontinuous at

$$t = \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}.$$

Since

$$\frac{\pi}{2} < 2 < \frac{3\pi}{2},$$

the largest open interval containing the initial point on which both coefficients are continuous is

$$\left(\frac{\pi}{2}, \frac{3\pi}{2}\right).$$

By Theorem 2.4.1, the solution is certain to exist throughout this interval.

□

Problem 2.4.5. State where in the ty -plane the hypotheses of Theorem 2.4.2 are satisfied for

$$y' = \sqrt{4 - t^2 - y^2}.$$

Solution. Let

$$f(t, y) = \sqrt{4 - t^2 - y^2}.$$

Then

$$f_y(t, y) = -\frac{y}{\sqrt{4 - t^2 - y^2}}.$$

The function f is real-valued and continuous where $t^2 + y^2 \leq 4$, while f_y is continuous only where

$$t^2 + y^2 < 4.$$

Hence both f and f_y are continuous in a rectangle about a point precisely when the point lies in the open disk

$$\boxed{t^2 + y^2 < 4}.$$

Thus the hypotheses of Theorem 2.4.2 are satisfied exactly in this region.

□

Problem 2.6.1. Determine whether

$$(4x + 3) + (6y - 1)y' = 0$$

is exact. If it is exact, find the solution.

Solution. Set

$$M(x, y) = 4x + 3, \quad N(x, y) = 6y - 1.$$

Then

$$M_y = 0 = N_x,$$

so the equation is exact. Let $\psi_x = M$ and $\psi_y = N$. Integrating $\psi_x = M$ with respect to x gives

$$\psi(x, y) = 2x^2 + 3x + h(y).$$

Therefore

$$\psi_y = h'(y) = 6y - 1,$$

so

$$h(y) = 3y^2 - y.$$

Hence the solution is given implicitly by

$$\boxed{2x^2 + 3x + 3y^2 - y = c}.$$

□

Problem 2.6.2. Determine whether

$$(3x - y) + (x - 3y)y' = 0$$

is exact. If it is exact, find the solution.

Solution. Set

$$M(x, y) = 3x - y, \quad N(x, y) = x - 3y.$$

Then

$$M_y = -1, \quad N_x = 1.$$

Since $M_y \neq N_x$, the equation is

not exact

.

□

Problem 2.6.3. Determine whether

$$(6x^2 - 2xy + 4) + (6y^2 - x^2 + 2)y' = 0$$

is exact. If it is exact, find the solution.

Solution. Set

$$M(x, y) = 6x^2 - 2xy + 4, \quad N(x, y) = 6y^2 - x^2 + 2.$$

Then

$$M_y = -2x = N_x,$$

so the equation is exact. Let $\psi_x = M$ and $\psi_y = N$. Integrating $\psi_x = M$ with respect to x gives

$$\psi(x, y) = 2x^3 - x^2y + 4x + h(y).$$

Therefore

$$\psi_y = -x^2 + h'(y) = 6y^2 - x^2 + 2,$$

which gives

$$h'(y) = 6y^2 + 2, \quad h(y) = 2y^3 + 2y.$$

Hence the solution is given implicitly by

$$\boxed{2x^3 - x^2y + 4x + 2y^3 + 2y = c}.$$

□